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Original Research

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Integrating Information and Communication Technologies to Enhance High School Students' Research Capabilities

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Abstract

Integrating information and communication technologies into research curricula can help students become more proficient researchers. Students need strong research capabilities in the modern educational environment to handle the challenges of information access and analysis. This article reveals the results of a study that explored the use of Science and Technology Academic and Research-Based Openly Operated Kiosk Stations (STARBOOKS), a program developed by the Department of Science and Technology (DOST) for use in public schools in the Philippines. The study investigated the effect of this technology on 36 purposefully selected Grade 9 students working on science investigatory projects in the Science, Technology, and Engineering Program of a public national high school. The participants were divided into seven groups, and their work was evaluated prior to the use of STARBOOKS and after. To determine any change in group performance, the panel of evaluators used standardized rubrics. Researcher-developed survey questionnaires were also used to obtain feedback from the students. Data were analyzed using descriptive statistics, such as weighted mean and percentages. The results showed an increase in the mean scores on the initial and final research projects. The study also revealed positive feedback from the students on STARBOOKS learning sessions. The results suggest the potential benefit of using the STARBOOKS online learning platform with lower grade levels as they begin their research subjects. Additionally, the results provide guidance to educators, organizations, and policymakers regarding the value of using information and communication technologies to enable students to become competent researchers.

Keywords: *high school students, ICT integration, research capability, technology-based learning*

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Introduction

Information and communication technologies (ICTs) have become integral tools in modern education; they have enhanced both teaching and learning processes for the current generation (Cruz, 2020). Ihmeideh and Al-Maadadi (2018) noted that computers, the internet, and other electronic delivery devices, such as radios, televisions, and projectors, are all commonly used in today's classrooms. ICTs are also used to promote a culture of lifelong learning outside of traditional classroom settings (Lock et al., 2021). Self-paced learning modules, webinars, and online courses allow people to stay up to date with their abilities, ensuring a dynamic and adaptable workforce and aligning with the concept of continuous learning that enjoys broad international support.

Technological advancements have eliminated geographical limitations, enabling widespread access to educational resources. Technology facilitates remote learning and improves underrepresented groups' access to education (Bituin Dorado & Alvar Ortega-Dela Cruz, 2024). This inclusivity is in keeping with the fourth goal of the United Nations' 2030 Agenda for Sustainable Development, which stresses high-quality education for all (Adipat & Chotikapanich, 2022). ICTs broaden the scope of education by enabling access from anywhere in the world at any time. Ghavifekr and Rosdy's (2015) study emphasized technology-based teaching and learning as one of the essential components in future development that can help teachers and students become more globally competitive.

Several studies have found that integrating ICTs into the teaching and learning processes yields positive results. Research indicates a positive correlation between academic achievement and ICT use. The use of technology in the classroom has a positive effect on students' attitudes and mathematical achievement, according to a 2019 study by Higgins et al. Quimsing and Cruz (2024) demonstrated the benefit of technology-based reading applications in improving reading literacy levels among struggling readers. Likewise, De Leon and Ortega-Dela Cruz (2024) studied the effect of web-based instruction on improving students' academic performance in music and arts education. The interactive and multimedia features of ICTs enhance language retention and understanding by bringing learning to life (Ọladeji et al., 2023). ICTs also offer opportunities for individualized, tailored training that is personalized to each student's needs (Holmes et al., 2018). Technology integration enables teachers to adapt assessments and content delivery to meet a variety of student learning styles and aptitudes (Cardino & Cruz, 2020).

Research has indicated that integrating ICTs helps learners strengthen their critical thinking abilities (Kong, 2015). Because digital tools are interactive, students are motivated to study well and practice creative thought (Yang et al., 2022). According to Jeong and Hmelo-Silver (2016), integrating interactive technologies creates collaborative learning settings and motivates students to participate actively in engaging with the material and one another. Positive attitudes toward learning and enhanced student motivation have been linked to the integration of ICTs. According to a study by Sun et al. (2018), when educational content is offered through digital platforms, students show higher levels of interest and passion, creating a more favorable learning environment.

The integration of technology in education equips students with skills essential in the 21st century (Cruz, 2020), the sort required in a knowledge-based society. The development of research capability is crucial for students to navigate the complexity of the information era successfully; it is a cornerstone of both academic and professional success. Strong research capabilities are essential in today's classrooms to help students traverse the ever-changing and complicated information landscape. In order to prepare students for the demands of a constantly changing labor market, education must also include such essential components as digital literacy, coding, and data analysis (Dimitrakopoulou, 2022). Teachers and students both benefit from the use of ICTs in the classroom. Research also indicates that teachers who participate in technology-enhanced professional development have more confidence and are more effective at using ICTs in their teaching practices (George & Sanders, 2017; Georgiou & Ioannou, 2019).

Integration of technology in the teaching and learning processes can be seen in public national high schools in the Philippines in the use of the Department of Science and Technology's (DOST) Science and Technology Academic and Research-Based Openly Operated Kiosk Stations (STARBOOKS), a program begun in 2011. Its aims are to entice more Filipino students to enroll in science and technology courses, to encourage innovations and inventions, and to support entrepreneurship and research. STARBOOKS was the first Philippine science digital library; it contained a significant body of information on science and technology packaged in a digital format. It was rich with information from international sources, presented in text, video, and audio formats. Notably, the original kiosks could be accessed without an internet connection, which was very helpful for schools in remote areas. Additionally, STARBOOKS featured a very user-friendly interface that even digital immigrants could easily understand. Moreover, it not only promoted science and technology but entrepreneurship as well, through Tamang DOSTkarte Livelihood Videos (Department of Science and Technology Science Technology Information Institute, 2015).

The efforts of the DOST STARBOOKS program led to the successful experience of a national high school in Bulacan Province, where the STARBOOKS installation was very helpful in increasing the performance of the students in the National Achievement Test. As a result, DOST went on to install more STARBOOKS all over the Philippines (Cinco, 2016). On September 29, 2016, a public national high school in Laguna province was awarded the 1000th STARBOOKS, a timely event, since the school had begun the division-initiated Science Technology and Engineering (STE) Program in 2015.

By 2021, however, Palines and Cruz, studying the research capabilities of STE students at a national high school, found that STARBOOKS was the least utilized learning resource in the school. Despite earlier interest in using STARBOOKS to improve science investigations, these unfortunate findings revealed that the units in the school's library were no longer functioning. Meanwhile, in the STARBOOKS webinar series conducted in July 2020, entitled Reframing STEM Education in the New Normal, the resource speaker announced that the DOST STARBOOKS would henceforth be accessed through online applications and websites (Department of Science and Technology Science and Technology Information Institute, 2015). Therefore, we took the opportunity to utilize the DOST STARBOOKS online learning platform to improve the science investigatory projects (SIPs) of Grade 9 students taking Research III.

Purposes of the Study

Developing SIPs, wherein students write and present their own science investigations, had been observed to be especially difficult amongst Grade 9 students. The study conducted by Palines and Cruz (2021) proved how hard students were struggling in research subjects; they found students to have low proficiency in writing and presenting their research in general. With this in mind, we performed a follow-up study that explored the use of DOST STARBOOKS as an intervention, focusing on helping students find reputable research and ideas in developing their SIPs. Hypothesizing that the use of this intervention would facilitate the teaching and learning of research subjects, we investigated the use of DOST STARBOOKS in improving the students' research capability, as demonstrated in their SIP proposals. The results of this study will be used in enhancing teaching strategies and providing standard learning resources for developing SIPs, with an eye toward improved student capability in writing research papers.

Specifically, the study determined students' awareness of the existence of DOST STARBOOKS, analyzed the mean difference between the pre- and post-intervention research outputs, and examined the students' feedback in relation to their use of DOST STARBOOKS in completing their SIPs.

Methods

Research Design

This study used a mixed-methods research design, in particular a combination of case study and survey research. Case study involves an in-depth investigation of a contemporary, real-life phenomenon in its context (Priya, 2021), whereas survey research uses a list of questions to collect data about a group of people (Gaur et al., 2020). This study used the case of high school students taking the Research III course.

Participants

The respondents of the study were purposefully selected on the basis of their being students taking the Research III course. All the students from Grade 9 under the STE program at a public national high school were identified as participants in the study, a total of 36. They were divided into seven groups, and each group created a research proposal.

Instrumentation

To investigate the perceived knowledge of the students about DOST STARBOOKS, the study used a researcher-developed survey questionnaire through Google Forms. The survey also examined the current needs of the students in developing their SIPs. Based on the results of the survey and the initial evaluation of the research papers, we created an online training schedule on the proposed platform. Students' experience in using DOST STARBOOKS was also gathered through an online survey. After the intervention, the final SIP proposals were evaluated and compared to the initial research submitted.

For the study, we developed a survey instrument that used a 5-point Likert scale to determine which part of DOST STARBOOKS most helped students improve their SIPs. Subject matter experts, including two science teachers and one research teacher, validated the researcher-made instrument. The original questionnaire created by the researchers was somewhat modified as a result of this validation. To examine the results of the respondents' initial and final research paper scores, we compared the scores gathered from three evaluators using the holistic grading rubrics for SIPs (Department of Education, 2018).

Moreover, feedback from the students was gathered in interviews to improve the facilitation of the learning sessions.

Data Analysis

Descriptive statistics were used to analyze the data obtained from the survey. Basically, to determine the perceived awareness of students in using DOST STARBOOKS, frequencies and percentages were used. To analyze the differences in the mean score of the initial and final research papers, descriptive analysis was used. Also, the researchers used descriptive analysis guided by the Science Investigatory Project Proposal Criteria from the Division of Laguna based on Caintic and Cruz's (2008) *Scientific Research Manual*. We developed a scale from 1 to 4, with 3.01–4.00 representing Advanced, 2.01–3.00 representing Proficient, 1.01–2.00 representing Developing, and 0.00–1.00 representing Beginning.

Additionally, qualitative data taken from interviews with students were transcribed verbatim.

Ethical Considerations

Prior to the study, a permission letter was submitted to and approved by the principal, and students' assent and parents' consent were secured. All information submitted by the respondents was handled with the highest confidentiality, and data privacy was closely maintained. Moreover, the participants preserved their anonymity in their manuscripts.

Results

Following are the findings of the study: the students' perceived awareness of the DOST STARBOOKS, the mean scores of the initial research papers submitted before the training, the mean scores of the final research papers submitted after the training, and the mean differences between the results after the evaluation.

Figure 1 reveals the percentage of learners initially aware of the existence of the DOST STARBOOKS kiosk in the school. In this particular survey, only 30 of the 36 respondents participated. The results indicate that 60% of the learners knew that three STARBOOKS kiosks were in the library. However, even if the students were aware of the existence of the reference materials, only 16.7% had created an account. A few of the learners also mentioned that they were able to use the kiosk in doing their assignments in history and research subjects.

Figure 1. *Percentage of Respondents' Perceived Awareness of STARBOOKS' Existence*

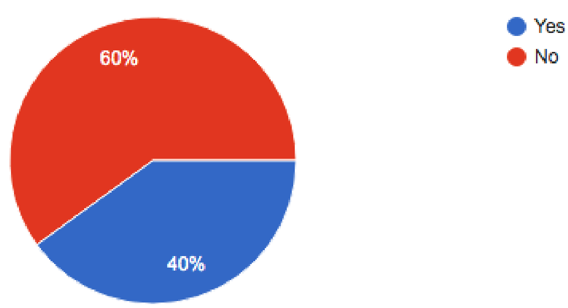

Table 1 shows the scores during the initial evaluation of students' research papers. The overall mean of 1.44 indicates that, on average, the research papers were at the Developing level. Among the seven research titles, four are Developing, one is Proficient and two are at the Beginning level, which suggests the participants require more help in writing their SIPs. The team who proposed the study Lemon Grass Extract as an Additive Component in DIY Hydroponic Solution suggests a great need for improvement, with a mean of 0.08 (Beginning), whereas the students who proposed Debarment of Tin and Lead Using Cigarette Butts appear to have successfully integrated their prior learning, with a mean of 2.22 (Proficient). However, the results indicate that only 14% of these Grade 9 research students have a clear understanding of the contents needed in writing an SIP proposal.

Table 1. *Mean Scores for Research Proposals Before Online Training*

Research title	Mean	Descriptive interpretation
Arduino Information and Body Temperature Scanner	1.68	Developing
Lemon Grass (*Cymbopogon citratus*) Extract as an Additive Component in DIY Hydroponic Solution	0.08	Beginning
Automated Hydro-Powered Mill as an Alternative Energy Source	1.82	Developing
Debarment of Tin (Sn) and Lead (Pb) Using Cigarette Butts	2.22	Proficient
Arduino Earthquake Detector for Quick Signaling to Authorities	0.98	Beginning
Mango (*Mangifera indica*) Peel and Eggshell as an Alternative Fertilizer for Okra (*Abelmoschus esculentus*)	1.70	Developing
Garlic (*Allium sativum*) and Oregano (*Oregano vulgare*) Extract as an Alternative Candle Mosquito Repellent	1.63	Developing
Overall mean	1.44	Developing

Note: Range: 0–1.00 = Beginning, 1.01–2.00 = Developing, 2.01–3.00 = Proficient, 3.01–4.00 = Advanced.

Certainly, students differ greatly from one another, and academic diversity within the student body is increasing. Accordingly, adjustments to certain aspects of the teaching and learning processes are required to guarantee that every student in the classroom has an equal opportunity to learn (Tomlinson, 2017). These circumstances acted as a stimulus for implementing interventions to improve student performance.

Table 2 presents the final research paper scores of the Grade 9 participants, showing their performance after the utilization of the DOST STARBOOKS online learning platform. It exhibits an overall mean of 3.06, which indicates that, in general, students have reached the Advanced level in writing their papers following the intervention. Of the seven research titles, four were evaluated as Advanced, and three were evaluated as Proficient. The team with the SIP title Microbial Fuel Cell from Tilapia (*Oreochromis niloticus*) Scale Phosphate Soluble Bacteria ranked first, with a mean of 3.53, whereas the team that shifted from Arduino Earthquake Detector for Quick Signaling to Authorities to Calamansi as an Alternative Laundry Detergent Product rated as Proficient, with a mean of 2.50.

Table 2. *Mean Scores for Research Proposals After Online Training*

Research title	Mean	Descriptive interpretation
Arduino Information and Body Temperature Scanner	2.98	Proficient
Lemon Grass (*Cymbopogon citratus*) Extract as an Additive Component in DIY Hydroponic Solution	3.33	Advanced
Automated Hydro-Powered Mill as an Alternative Energy Source	2.53	Proficient
Microbial Fuel Cell from Tilapia (*Oreochromis niloticus*) Scale Phosphate Soluble Bacteria	3.53	Advanced
Calamansi (*Citrofortunella microcarpa*) as an Alternative Laundry Detergent Product	2.50	Proficient

Mango (*Mangifera indica*) Peel and Eggshell as an Alternative Fertilizer for Okra (*Abelmoschus esculentus*)	3.43	Advanced
Garlic (*Allium sativum*) and Oregano (*Oregano vulgare*) Extract as an Alternative Candle Mosquito Repellent	3.15	Advanced
Overall Mean	3.06	Advanced

Note: Range: 0–1.00 = Beginning, 1.01–2.00 = Developing, 2.01–3.00 = Proficient, 3.01–4.00 = Advanced.

The results support the findings of the study by Sarode (2018), which emphasizes the importance of teacher styles and strategies in the learning process. The students' levels of performance were productive for diverse types of learners. Based on student feedback, they were able to produce positive results in their own way, suggesting the effectiveness of the teaching approach

Table 3 shows the differences between the mean scores of the students' evaluated pre- and post-training research proposals. There is an increase from the initial scores to the final scores in all the research proposals submitted by the teams. The team with the project entitled Lemon Grass Extract as an Additive Component in DIY Hydroponic Solution showed the greatest improvement, with the highest mean difference of 3.25, coming in third place among the seven teams. The team with the project entitled Automated Hydro-Powered Mill as an Alternative Energy Source had the least mean difference value of 0.71, progressing from the Developing level to Proficient level. The results display the improvement in students' performance after the use of DOST STARBOOKS.

Table 3. *Mean Differences Between Scores for Research Proposals*

Research title	Mean scores of the research proposals		Mean differences
	Initial	Output	
Arduino Information and Body Temperature Scanner	1.68	2.98	1.30
Lemon Grass (*Cymbopogon citratus*) Extract as an Additive Component in DIY Hydroponic Solution	0.08	3.33	3.25
Automated Hydro-Powered Mill as an Alternative Energy Source	1.82	2.53	0.71
Microbial Fuel Cell from Tilapia (*Oreochromis niloticus*) Scale Phosphate Soluble Bacteria	2.22	3.53	1.31
Calamansi (*Citrofortunella microcarpa*) as an Alternative Laundry Detergent Product	0.98	2.50	1.52
Mango (*Mangifera indica*) Peel and Eggshell as an Alternative Fertilizer for Okra (*Abelmoschus esculentus*)	1.70	3.43	1.73
Garlic (*Allium sativum*) and Oregano (*Oregano vulgare*) Extract as an Alternative Candle Mosquito Repellent	1.63	3.15	1.52
Overall mean	1.44	3.06	1.62

Note: Range: 0–1.00 = Beginning, 1.01–2.00 = Developing, 2.01–3.00 = Proficient, 3.01–4.00 = Advanced.

To support the data presented on the improvement in students' scores and its connection to the use of STARBOOKS, the participants in the study completed the feedback survey, which asked if they really used the online learning platform for references in their SIPs. The results show that 100%, or 35 out of 35 respondents, confirmed their usage of STARBOOKS. In terms of usage frequency, 60%, or 21 out of 35 respondents, answered often, while 20%, or 7 of 35 respondents, confirmed that they always use STARBOOKS. Worth noting is the fact that nobody answered that they never used STARBOOKS.

Student participants were also asked about their level of satisfaction with the use of DOST STARBOOKS and 57.1%, or 20 of 35 respondents, said they were highly satisfied. Further, 68.6% (24) indicated they will still use the DOST STARBOOKS online learning platform for references in their future scientific studies. The respondents said that they would highly recommend the digital library. Furthermore, 82.9% of respondents agreed that they would highly recommend the platform as a source for research.

Following are some of the students' key takeaways and realizations from the learning sessions:

"We highly needed this kind of learning, since it is pandemic right now, and it is very helpful in conducting their research." (Participant 1)

"It is good to attend online training, because the training itself will help you increase your knowledge about something that you do not know yet." (Participant 2)

"It is much easier when we are using DOST STARBOOKS, because all the things that we need in doing research is inside of this." (Participant 3)

"DOST STARBOOKS is a good platform for students or researchers to get some ideas and references that can help them." (Participant 4)

"DOST STARBOOKS is very helpful for most researchers; they can find references and journals that are connected with their studies." (Participant 5)

"There are so many things that can be found in the internet that are free and reliable. One of them is the DOST STARBOOKS." (Participant 6)

"DOST STARBOOKS is very useful, and I have learned a lot from the training." (Participant 7)

The students' comments indicate how appreciative they were of the intervention's execution because, all things considered, it greatly aided their academic endeavors. Some students did, however, note that they were not all that interested because they also needed to learn the platform. Additionally, students requested that games be included in the sessions to improve their relationships with teachers and fellow students.

Discussion

ICTs play a crucial role in enhancing student engagement, as demonstrated by numerous studies. When integrated into the curriculum, technology has been shown to boost student's interest, participation, and motivation (Haleem et al., 2022). Platforms that foster student engagement highlight the collaborative advantages of technology, promoting teamwork, knowledge-sharing, and a sense of community (Jeong & Hmelo-Silver, 2016). ICTs also support communication and collaboration within educational communities, contributing to a more cohesive and supportive learning environment. Tools for communication and online resources enhance interactions between parents, teachers, and students (Livingstone, 2015).

Moreover, studies consistently show a positive correlation between the use of ICTs and improved student achievement, as noted earlier. For instance, Higgins et al. (2019) demonstrated that technology use enhanced students' attitudes and performance in mathematics. The integration of ICTs also increases the flexibility of the learning process. Holmes et al. (2018) found that students appreciated the ability to access resources, collaborate with peers, and complete assignments at their own pace. Furthermore, citation management systems are valuable tools that help students organize references, create bibliographies, and adhere to proper citation guidelines during research (Mantikayan & Abdulgani, 2017).

The integration of ICTs into education has far-reaching implications for both teaching and learning. On one hand, it empowers students to take ownership of their learning by providing greater flexibility and control over how and when they access resources. This flexibility not only caters to diverse learning styles but also encourages self-paced learning, which can boost motivation and academic performance. On the other hand, the use of technology in classrooms necessitates continuous training and professional development for educators to effectively incorporate ICT tools into their teaching strategies.

Additionally, the collaborative nature of many ICT platforms fosters a more interactive and dynamic learning environment. As students engage with their peers and teachers through digital platforms, they develop vital teamwork and communication skills that are crucial in today's digital and interconnected world.

Conclusion

This study investigated the possibility that incorporating ICTs could aid students in developing their research capabilities. Specifically, this article explores how the DOST STARBOOKS were used to improve the skills of Grade 9 students in a public national high school in the Philippines in developing their SIPs for the Science, Technology, and Engineering Program.

Based on the results of the study, we conclude that DOST STARBOOKS learning sessions increased students' performance in terms of writing their SIP proposals. The findings of the study also show that prior to the study, only a few students knew about DOST STARBOOKS in the school, and they did not use the resource fully. Through the efforts of the study, all the respondents were made aware of the online learning platform and were able to integrate the use of DOST STARBOOKS in their own research after the intervention sessions. Respondents also shared positive experiences throughout the span of the study.

The lowest mean scores from the panelists' evaluation, based on the indicators or contents of each proposal, should be taken into account when developing future intervention activities for learners. Specifically, science teachers should undergo capacity-building programs to effectively use the DOST STARBOOKS online learning platform. This will enhance their ability to integrate the platform into their teaching practices, making it easier for science and research teachers to utilize DOST STARBOOKS in their classrooms.

This study suggests that to boost student participation during class, teachers should use games and activities. For educators to stay current with emerging technologies and educational approaches, they must engage in ongoing professional development. Also, effective technology integration into research education requires educators to stay up to date on the newest pedagogical approaches and technologies.

The use of the DOST STARBOOKS online learning platform may also be introduced to lower grade levels as they begin their research subjects. Learning tasks involving the use of DOST STARBOOKS may be integrated into both science and research classes. To reinforce the learning, students may develop their starting SIPs using DOST STARBOOKS references. Moreover, multiple virtual training courses with scientists and researchers are highly recommended.

The integration of technology has the potential to improve students' research capabilities. This article presents evidence that highlights the revolutionary impact of ICT on research processes, collaboration, and resource access. Teachers and educational institutions are encouraged to implement these strategies and recommendations to enable students to become competent researchers in the digital era.

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